

A Proof of Lallouet's Conjecture on OEIS A059324, with a Converse

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August 2026

Abstract

Let [A059324](#) denote the sequence of positive integers n for which $6n + 5$ is composite. In March 2008, P. Lallouet conjectured that for such n there exists no pair of primes (p, q) with $q > p^2$ and $q - p^2 = 6n - 4$. We prove this conjecture. In fact we prove the sharper statement that such a pair exists *if and only if* $6n + 5$ is prime, in which case the pair is unique in its first coordinate, namely $p = 3$ and $q = 6n + 5$. Thus [A059324](#) is precisely the set of $n \geq 1$ for which no such pair exists. The proof is elementary, resting on the observation that every prime $p \geq 5$ satisfies $p^2 \equiv 1 \pmod{6}$, which forces $3 \mid q$ and hence $q = 3$, contradicting $q > p^2$.

2020 Mathematics Subject Classification. 11A41, 11A07, 11B83.

Keywords. prime numbers, quadratic residues modulo 6, OEIS A059324, Dirichlet's theorem.

1 Introduction

The sequence [A059324](#) of the On-Line Encyclopedia of Integer Sequences [1], contributed by Anton Joha in January 2001, is defined by

$$\text{A059324} = \{ n \geq 1 : 6n + 5 \text{ is composite} \},$$

with initial terms

5, 10, 12, 15, 19, 20, 23, 25, 26, 30, 33, 34, 35, 36, 40, 45, 47, 49, 50, 53, ...

For instance 12 is a term because $6 \cdot 12 + 5 = 77 = 7 \cdot 11$. Its complement in the positive integers is [A059325](#), the set of n with $6n + 5$ prime.

On 20 March 2008, Philippe Lallouet contributed the following comment to the entry:

*Conjecture: There exists no pair of primes $(p, q > p^2)$ such that $q - p^2 = 6 * n - 4$ (see [A138479](#)).*

Two readings of the symbol n are conceivable, and we record at once which is intended. Under the *index* reading, n would be the position of a term and the assertion would range over all $n \geq 1$; this reading is untenable, since $n = 1$ gives $6n - 4 = 2$ and the primes $(p, q) = (3, 11)$ satisfy $11 > 9 = 3^2$ and $11 - 9 = 2$, falsifying the statement immediately. Under the *value* reading, n is a term of [A059324](#), so that the assertion is made precisely for those n with $6n + 5$ composite. We adopt the value reading throughout; it is the only one under which the conjecture is not trivially false, and it is the standard convention for comments attached to a sequence of this type. The cross-reference to [A138479](#) in the comment points to a related sequence and plays no role in what follows.

At the time of writing the statement still stands on the entry as an unproven conjecture.

The purpose of this note is to prove it. We in fact establish a complete characterisation, of which Lallouet's conjecture is one direction.

Theorem 1. *Let $n \geq 1$ be an integer. There exist primes p and q with*

$$q > p^2 \quad \text{and} \quad q - p^2 = 6n - 4$$

if and only if $6n + 5$ is prime. Moreover, when such a pair exists it satisfies $p = 3$ and $q = 6n + 5$, and no other value of p is possible.

Corollary 2 (Lallouet's conjecture). *If $n \in$ [A059324](#), that is if $6n + 5$ is composite, then there exists no pair of primes (p, q) with $q > p^2$ and $q - p^2 = 6n - 4$.*

Corollary 3. *For $n \geq 1$, no such pair exists precisely when $6n + 5$ is composite. Hence [A059324](#) is exactly the set of $n \geq 1$ for which no such pair exists, its complementary condition “ $6n + 5$ prime” (recorded as [A059325](#)) being exactly the case in which one does.*

Corollary 3 shows that Lallouet's conjecture is not merely true but sharp: the hypothesis “ $6n + 5$ composite” cannot be weakened, since every n omitted from [A059324](#) admits an explicit pair.

2 Preliminaries

We use only the following elementary observation.

Lemma 4. *If p is a prime with $p \geq 5$, then $p^2 \equiv 1 \pmod{6}$.*

Proof. Since $p \geq 5$ is prime, p is divisible by neither 2 nor 3, so $\gcd(p, 6) = 1$ and hence $p \equiv 1$ or $5 \pmod{6}$. In the first case $p^2 \equiv 1 \pmod{6}$; in the second, $p^2 \equiv 25 \equiv 1 \pmod{6}$. $\square$

Remark 5. Equivalently, $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ is a group of order 2, so every element squares to the identity.

3 Proof of Theorem 1

Proof. Fix $n \geq 1$ and set $d := 6n - 4$, so that $d \geq 2$ and $d \equiv 2 \pmod{6}$. We seek primes p, q with $q = p^2 + d$ and $q > p^2$; the latter is automatic since $d > 0$. We examine the three possible cases for p separately.

Case 1: $p \geq 5$. By Lemma 4, $p^2 \equiv 1 \pmod{6}$, and therefore

$$q = p^2 + d \equiv 1 + 2 \equiv 3 \pmod{6}.$$

In particular $3 \mid q$. Since q is prime, this forces $q = 3$. But $q > p^2 \geq 25$, a contradiction. Hence no solution has $p \geq 5$.

Case 2: $p = 2$. Here $p^2 = 4$ and

$$q = 4 + (6n - 4) = 6n.$$

For $n \geq 1$ we have $6n \geq 6$ and $6 \mid 6n$, so q is composite. Hence no solution has $p = 2$.

Case 3: $p = 3$. Here $p^2 = 9$ and

$$q = 9 + (6n - 4) = 6n + 5.$$

The condition $q > p^2$ reads $6n + 5 > 9$, which holds for every $n \geq 1$. Therefore a solution with $p = 3$ exists if and only if $6n + 5$ is prime, and in that case $q = 6n + 5$ is determined.

Cases 1 and 2 exclude every prime $p \neq 3$. Case 3 shows that $p = 3$ yields a solution exactly when $6n + 5$ is prime, with q then uniquely determined. This proves both directions of the equivalence and the stated uniqueness. $\square$

Proof of Corollary 2. If $n \in \text{A059324}$ then $6n + 5$ is composite by definition, so by Theorem 1 no pair exists. $\square$

Proof of Corollary 3. Immediate from Theorem 1: for each $n \geq 1$ exactly one of “ $6n+5$ composite” and “ $6n + 5$ prime” holds, and the theorem attaches non-existence to the former and existence to the latter. $\square$

4 Sharpness and the excluded values

The following makes explicit how far the conjecture’s hypothesis is from removable.

Proposition 6. *For every $n \notin \text{A059324}$ the pair $(p, q) = (3, 6n + 5)$ satisfies all the required conditions, and there are infinitely many such n .*

Proof. If $n \notin \text{A059324}$ then $q := 6n + 5$ is prime, $q > 9 = 3^2$, and $q - 3^2 = 6n - 4$, so the pair qualifies.

For the infinitude, it suffices to show there are infinitely many primes $q \equiv 5 \pmod{6}$, since each such $q > 9$ is $6n + 5$ for a unique $n \geq 1$. We give the elementary argument rather than appeal to Dirichlet’s theorem. Suppose the primes congruent to 5 modulo 6 were finitely many, say $q_1, \dots, q_k$ (the list is nonempty, as 5 is one). Put

$$N := 6 q_1 q_2 \cdots q_k - 1.$$

Then $N > 1$ and $N \equiv -1 \equiv 5 \pmod{6}$. Since N is odd and not divisible by 3, every prime factor of N is coprime to 6 and hence congruent to 1 or 5 modulo 6. Were every prime factor congruent to 1, their product N would be congruent to 1 modulo 6, contradicting $N \equiv 5$. So some prime factor q satisfies $q \equiv 5 \pmod{6}$. But $q \neq q_i$ for every i : otherwise q_i would divide both N and $6q_1 \cdots q_k$, hence divide their difference 1, which is absurd. This contradicts the assumed completeness of the list. $\square$

The smallest excluded value is $n = 1$, giving $d = 2$ and the pair $(3, 11)$: indeed 11 and 3 are prime, $11 > 9$, and $11 - 9 = 2$. The next few are

$(3, 17), (3, 23), (3, 29), (3, 41), (3, 47), (3, 53), (3, 59), (3, 71), (3, 83), (3, 89), \dots$

corresponding to $n = 2, 3, 4, 6, 7, 8, 9, 11, 13, 14, \dots$, none of which lies in [A059324](#).

Remark 7. The prime 3 is exceptional here for a structural reason: it is the unique prime p with $p^2 \equiv 3 \pmod{6}$. For $p = 2$ one has $p^2 \equiv 4$, and for all $p \geq 5$ one has $p^2 \equiv 1$ by Lemma 4. Since $d \equiv 2 \pmod{6}$, the residue $p^2 + d \pmod{6}$ equals 0, 5, 3 according as $p = 2$, $p = 3$, $p \geq 5$; and of these only 5 is coprime to 6. The whole theorem is this single table.

5 Computational verification

The results above are proved, so computation serves only as a guard against error in their statement. We report the checks carried out.

- The 62 listed terms of [A059324](#) were confirmed to satisfy “ $6n + 5$ composite”, and the list was confirmed to be exactly $\{n \leq 130 : 6n + 5 \text{ composite}\}$.
- For every listed term n and every prime $p < 200$, the number $p^2 + 6n - 4$ was tested for primality. No pair satisfying the conditions was found, in agreement with Corollary 2.
- For each of the 68 values $n \leq 130$ with $n \notin \text{A059324}$, the pair $(3, 6n + 5)$ was confirmed to satisfy all conditions, in agreement with Corollary 3.
- Lemma 4 was verified for all 9590 primes p with $5 \leq p < 10^5$: no exceptions.

6 Concluding remarks

The argument is elementary and uses nothing beyond the structure of $(\mathbb{Z}/6\mathbb{Z})^\times$. We make no claim as to its difficulty. Its interest lies in the fact that the conjecture, recorded in 2008, admits a complete resolution in a few lines once one asks not whether solutions are scarce but which prime p could possibly occur; the congruence condition then eliminates all but one candidate outright.

We have been unable to locate this argument in the literature, but it is short enough that prior knowledge would not be surprising, and we would welcome notice of any earlier appearance.

References

- [1] N. J. A. Sloane et al., The On-Line Encyclopedia of Integer Sequences, <https://oeis.org>, 2026. Sequences [A059324](#) (numbers n such that $6n + 5$ is composite), [A059325](#) (numbers n such that $6n + 5$ is prime).